

Data Wrangling 🤠

Variables

Lesson 4

Variables

- **Variables** allow us to bind a name to objects in memory
- We can *store*, *read*, and *change* values in a variable
- Variable initialization denoted by: `[variable_name] = [value]`
- For example: `x = 7`

Naming Conventions

- Identifiers (i.e. variable names) must begin with a letter or underscore and cannot contain spaces
- **snake_case**: separate words in a variable with underscores and keep all characters lowercase
- For example: `multi_word_variable_name`

Variable Assignment Breakdown

1. Evaluate the RHS: Compute the right-hand side expression
2. Assign to the LHS: Store the result in the variable named on the left-hand side

For example: `x = 2 + 3` → evaluate `2 + 3` (get `5`) → set `x` to `5`

Demo: Creating & Using Variables

```
>>> x = 7
>>> x
7
>>> x + 3
10
>>> x - 3
4
>>> x = 10
>>> x
10
>>> x = 10 * 3
>>> x
30
```

Demo: Many Variables

```
>>> y = 10
>>> z = 5
>>> x + y + z
45
>>> x = y
>>> x
10
>>> x == y
True
>>> x = x + 1
>>> x == y
False
>>> x
11
```

Demo: Variables with Strings

```
>>> x = "hello"
```

```
>>> y = "class"
```

```
>>> x == y
```

```
False
```

```
>>> x + " " + y
```

```
"hello class"
```

```
>>> x = x + "!"
```

```
>>> x
```

```
"hello!"
```